Lakshmi Prasanna Sriram Appasani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	90.90%
Sri Chaitanya High School	Class X (Board of Secondary Education, Andhra Pradesh)	2021	100.00%

SKILLS

Languages: C, C++, Python | **Frontend:** HTML, CSS, JavaScript, React, Tailwind CSS, Bootstrap

Backend & Databases: Node.js, Express.js, MySQL, MongoDB

Libraries & Tools: NumPy, Pandas, Scikit-learn, Streamlit, Matplotlib, Git, GitHub, Solid Edge

Competitive Programming: Solved 400+ LeetCode Problems

RELEVANT COURSES

- **Computer Science:** Data Structures and Applications, Algorithms, Computer Networks, Software Engineering, Database Management Systems, Operating Systems, Theory of Computation,Discrete Maths, Introduction to Computing, Introduction to Programming
- **Artificial Intelligence:** Machine Learning, Fraud Analytics,AI for Finance
- **Mathematics:** Convex Optimisation, Probability and Random Variables, Number Theory, Linear Algebra, Statistics, Calculus, Differential Equations, Transform Techniques, Complex Variables

PROJECTS

Spatio-Temporal Action Recognition

Computer Vision, Deep Learning, Video Classification, Temporal Modeling

- Built a **frame-based CNN action classifier** on UCF101, establishing a spatial-only baseline without modeling temporal dependencies.
- Enhanced action recognition by feeding **CNN frame features into an LSTM** to capture motion and temporal context, achieving **~90% accuracy** and demonstrating the advantage of sequence-aware modeling.
- Applied **t-SNE to last-layer features** to visualize class-wise representation quality and temporal feature separation.

AI-Powered Adaptive Authentication & IAM Gateway

Tech: Node.js, Express, MongoDB Atlas, React, JWT, Passport.js

- Built a centralized **SSO/MFA gateway** serving **5,000+ students** across institute-wide applications, supporting per-app configurable auth methods (**local, OAuth2, SAML2, OIDC**) with **MongoDB TTL-backed session management**.
- Designed a **risk-scoring model** using login behavior, geolocation, and **VPN/proxy detection** to dynamically bypass MFA for low-anomaly sessions and enforce step-up verification for high-anomaly ones.
- Engineered auth security infrastructure with **JWT access/refresh tokens**, **bcrypt hashed credential storage**, scheduled key rotation with backward validation, and **CIDR-based IP whitelisting**.
- Implemented **immutable audit logging** across **13+ security event types** to support traceability and post-incident review.

VMDT - Verifiable Multipath Decentralized Transport Protocol

C++, Python, TCP/UDP, Computer Networks, Cryptography, P2P Systems

- Designed a **decentralized multipath transport protocol** for resilient **P2P communication**, distributing data across parallel network paths.
- Integrated **Shamir's Secret Sharing** with **AES-256-CTR encryption** for secure data distribution, ensuring **confidentiality and fault tolerance** under node/path failures.
- Built a **dual-layer TCP/UDP architecture** for **peer discovery, cluster management, and multipath transfer**.
- Implemented a **Reliable Data Transfer (RDT)** layer with **acknowledgments and retransmissions** for reliable UDP communication.

Radiomics MRI Analysis Web Application

Python, SimpleITK, PyRadiomics, Scikit-learn, Flask/React, DICOM/NIfTI, Multithreading

- Built an **end-to-end MRI preprocessing pipeline (denoising, N4 bias-field correction, normalization, coregistration, skull-stripping)** with a **DICOM-to-NIfTI conversion** module to standardize scans for analysis.
- Used **radiomics and ML** for automated **tumor identification**, achieving **[~90]% accuracy/AUC on [500] scans.**
- Used **multithreading** to offload compute-heavy operations, ensuring a **responsive, non-blocking UI** during scan processing.

EXTRACURRICULAR

Internship Coordinator, Office of Career Services (OCS)

- Coordinated internship-related activities for **600+ students**, facilitating communication between **students, recruiters, and the placement team** and supporting internship-related coordination.

Active Member, NSS-IITH

- Participated in **community service, social outreach, and institute-level volunteering activities**.

Sports Activities

- Actively participated in **volleyball and cricket** through **institute-level leagues**, contributing to team-based sporting activities.